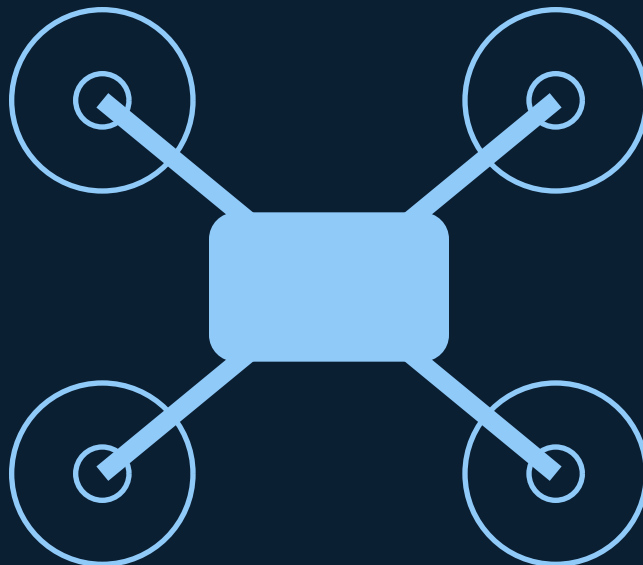


Jakota Parker

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Custom 3-Inch FPV Quadcopter

End-to-end hardware design, mechanical integration, electronics planning, manufacturing, and flight-controller validation.



PROJECT FOCUS

Parametric CAD

Electronics Integration

Additive Manufacturing

Betaflight

Design Verification
[GitHub repository](#)

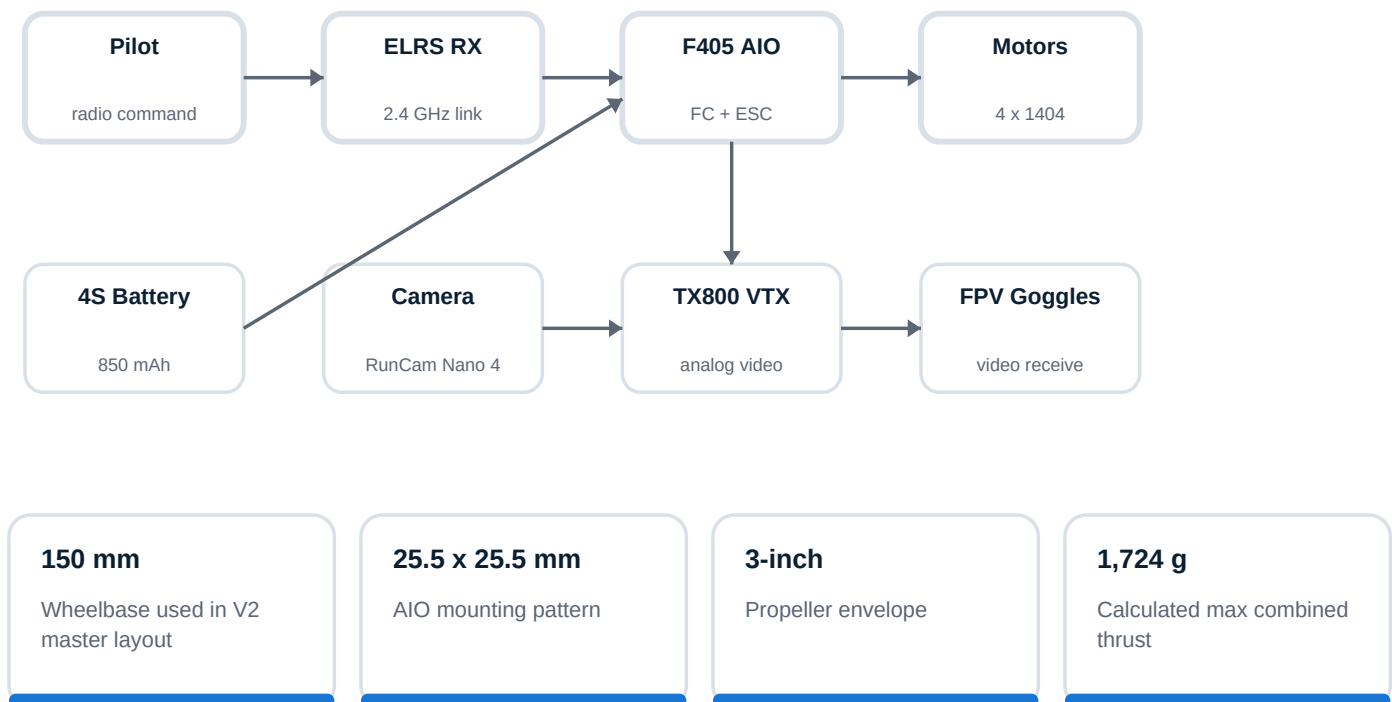
Scan for full documentation



Project Architecture

A compact aircraft treated as a complete electromechanical system, not just a frame.

Objective. Design, manufacture, assemble, configure, and test a custom 3-inch FPV quadcopter using a primarily 3D-printed airframe, while maintaining professional engineering documentation across CAD, electronics, manufacturing, firmware, and testing.

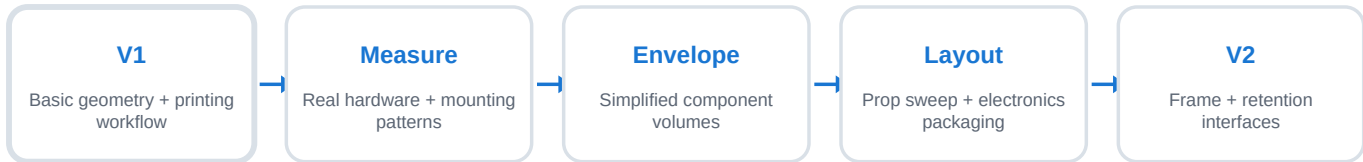


Key engineering decisions

- Built the V2 architecture around the actual purchased electronics rather than forcing parts into a pre-existing frame.
- Used component envelopes and propeller sweep regions to perform packaging and interference checks before printing.
- Centralized major dimensions in an Onshape Variable Studio so changes propagate through the model.
- Kept the battery near the center of mass, the camera front-center, and the VTX rearward for shorter antenna routing.

Parametric CAD + Packaging

V2 evolved from a geometric prototype into a hardware-driven design.



Master parameter system

Wheelbase	150 mm
AIO mount	25.5 x 25.5 mm
Motor hardware	M2
Propeller diameter	3 in / 76.2 mm
Controlled items	arm width, center body, clearances

Component envelope models

- SpeedyBee 1404 motor
- SpeedyBee F405 AIO
- RunCam Nano 4
- Tattu 4S 850 mAh
- SpeedyBee TX800 VTX
- BETAFPV ELRS Nano

Design-for-reality iteration

A major lesson was that nominal dimensions are not enough. Purchased components were measured with calipers and compared against CAD envelopes. Motor mounting geometry, screw length, USB access, camera clearance, VTX placement, receiver retention, battery support, and wire paths were treated as physical constraints to validate before flight.

Mechanical verification

Caliper-based dimensions, hole spacing, clearance checks, fit testing, and print iteration.

Manufacturing

Endeavor 3 V2 prototyping with PLA+ for the frame and TPU for compliant retention parts.

Revision control

Separate V1/V2 design revisions, changelog discipline, and exported STL/STEP artifacts.

ENGINEERING PRINCIPLE

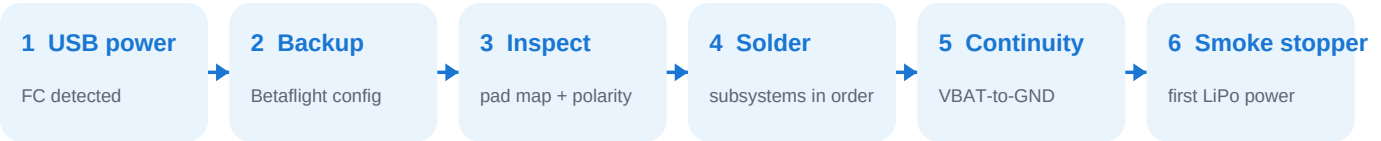
Model what matters, measure what is uncertain, and keep geometry parametric enough to absorb change. This became the core V2 workflow.

Electronics Integration

Power, signaling, pad mapping, safe bring-up, and subsystem verification.

Subsystem	Selected hardware	Interface	Engineering concern
Flight controller	SpeedyBee F405 AIO V2	USB / UART / video / ESC	pad map, orientation, power rails
Receiver	BETAFPV ELRS Nano	UART TX/RX	protocol, UART selection, antenna
Camera	RunCam Nano 4	power + analog video	voltage, lens/cable clearance
Video TX	SpeedyBee TX800	power + video + control	voltage, antenna routing, heat
Propulsion	4 x 1404 4600KV	3-phase motor outputs	motor order, direction, screw safety
Power	4S 850 mAh LiPo	VBAT + GND	polarity, capacitor, short protection

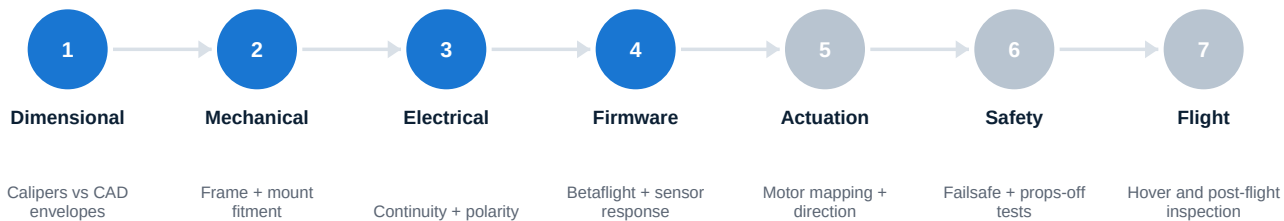
Safe bring-up sequence



Bench-test evidence: the AIO powered over USB, enumerated as a COM device, connected successfully to Betaflight Configurator, and the accelerometer was calibrated before soldering progressed.

Test, Debug, Iterate

The project is organized around evidence-based progression rather than “assemble and hope.”



Examples of engineering judgment

Fit before soldering Mock assembly and TPU retention checks happen before permanent electrical installation.

No props during setup Propellers stay removed through soldering, firmware configuration, receiver setup, and motor-direction testing.

Protect first power-on Visual inspection, polarity verification, continuity checks, and a smoke stopper precede first LiPo power.

Keep documentation live Routing or fit changes in hardware, the wiring/CAD documentation is updated to match the physical system.

Current build status

Mechanical packaging, TPU mounts, and electronics bench validation are complete enough to move into soldering and subsystem integration. The next milestones are full wiring, continuity/smoke-stopper checks, receiver/VTX setup, motor validation, failsafe testing, and first controlled hover.

What This Project Demonstrates

A small aircraft used as a platform for disciplined hardware engineering.

Capabilities demonstrated

- **System architecture** - decomposed command, power, video, propulsion, and structural subsystems.
- **CAD & packaging** - built parametric layouts and component envelopes around real hardware.
- **Electrical integration** - mapped power rails, UARTs, video signals, motor outputs, and safe soldering order.
- **Manufacturing** - iterated PLA+ frame and TPU retention parts on an Ender 3 V2.
- **Verification** - used measurement, fit checks, continuity, staged power-on, and Betaflight validation.
- **Technical documentation** - maintained architecture, BOM, build logs, test plans, CAD revisions, and electronics checklists in GitHub.
- **Debugging mindset** - treated discrepancies as inputs for redesign rather than one-off fixes.
- **Growth path** - using this platform as a foundation for

WHY THIS PROJECT MATTERS

This project forced me to work across mechanical and electrical boundaries: dimensions affect routing, routing affects maintainability, power choices affect thermal and safety constraints, and every physical change must still work with the flight-control stack. That systems-level interaction is the part of hardware engineering I want to keep developing professionally.

Full engineering documentation

GitHub: [JakotaParker / mini-fpv-drone](https://github.com/JakotaParker/mini-fpv-drone)

Includes CAD, build logs, BOM, electronics integration, manufacturing notes, firmware planning, and test documentation.

github.com/JakotaParker/mini-fpv-drone

